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Project 2
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Data Analysis

Joe's Stone Crab is a seafood restaurant famous for its fresh crabs and other seafood. In this project, we are trying to predict Joe's Stone Crabs' sales for 2024 with existing data of the 4 most recent years. We are given data from 2020 to 2023 and its sales, which are in thousands of dollars for units. For a majority of the data, the sales tend to increase as the year goes up for each month. This dataset was analyzed in several ways including finding average demand, seasonal factors, trend factors, and time series adjusting forecasting.

Season Factor

The seasonality in this project represents the recurring fluctuations in sales based on the time of year. The season factor for each month is calculated by dividing the average demand with the average of all the average demands in each month.

Month	2020 Sales (thousands)	2021 Sales (thousands)	2022 Sales (thousands)	2023 Sales (thousands)	Average Demand (thousands)	Season Factor	Busy or Slow
January	242	263	282	306	273	1.41	Busy
February	235	238	255	280	252	1.30	Busy
March	232	247	245	270	249	1.28	Busy
April	178	191	205	210	196	1.01	Busy
May	184	193	210	214	200	1.04	Busy
June	140	149	160	175	156	0.81	Slow
July	145	157	165	182	162	0.84	Slow
August	152	160	175	193	170	0.88	Slow
September	110	122	126	130	122	0.63	Slow
October	130	130	148	155	141	0.73	Slow
November	152	167	173	182	169	0.87	Slow
December	206	230	235	255	232	1.20	Busy
				Benchmark	193.42		

Figure 1. Season Factor

In **Figure 1**, January, February, and March have the most sales in the 4 year span and have the highest season factors with December right behind them. It may not be a coincidence that the months with the highest sales are right next to each other. This may be due to people wanting to celebrate holidays during that time period and deciding to spend their money to eat at the restaurant. Based on the season factor for each month through the 4 years, the restaurant is determined to be busy with a season factor of 1 or above. Any season factor that is less than 1 is considered to be slow. January, February, and March are going to be noticeably busy with their high season factor. September and October are going to be the least busy, as their seasonal factors are very low.

Trend Factor

The trend factor captures the long term growth pattern in Joe's Stone Crab monthly sales. To analyze the trend, a linear regression model was used to identify the general direction of sales over the 4 year period. The trend line equation was generated through statistical analysis using the regression model to find the intercept and the x variable. In **Figure 2**, we have the intercept as 169.89 and slope as .96, which we can just plug into the equation. The intercept or 169.89 represents the baseline level at the start of the analysis period. A slope of .96 suggests that the business has an increase of $960 per month. To find the forecast with trend for each month of every year, the period value was substituted for time. Once the equation was formed, the

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.931142424
R Square	0.867026214
Adjusted R Square	0.86413548
Standard Error	5.321438529
Observations	48

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	8493.410738	8493.411	299.932845	8.79886E-22
Residual	46	1302.614569	28.31771		
Total	47	9796.025307			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	169.8916368	1.560487762	108.8709	3.7654E-57	166.7505379	173.032736	166.750538	173.032736
X Variable 1	0.9602053	0.055443685	17.31857	8.7989E-22	0.848602951	1.07180765	0.84860295	1.07180765

FIT = 169.89 + 0.96*t

Figure 2 Regression Model

	37	306	1.41	217	205	290	15.81	249.84
	38	280	1.30	215	206	269	11.12	123.73
	39	270	1.28	210	207	266	3.62	13.14
	40	210	1.01	207	208	211	1.07	1.15
	41	214	1.04	207	209	217	2.64	6.98
2023	42	175	0.81	217	210	170	5.46	29.76
	43	182	0.84	217	211	177	4.86	23.59
	44	193	0.88	220	212	186	6.55	42.93
	45	130	0.63	206	213	134	4.41	19.44
	46	155	0.73	213	214	156	0.76	0.59
	47	182	0.87	209	215	187	5.31	28.21
	48	255	1.20	213	216	258	3.49	12.21
	49		1.41		217	306		
	50		1.30		218	284		
	51		1.28		219	281		
	52		1.01		220	223		
	53		1.04		221	229		
2024	54		0.81		222	179		
	55		0.84		223	187		
	56		0.88		224	197		
	57		0.63		225	142		
	58		0.73		226	164		
	59		0.87		227	197		
	60		1.20		227	272		

Figure 3 Forecast Table

forecast with trend and seasonality for 2024 was able to be calculated, even without knowing the demand. The forecast with seasonality was found by just multiplying the season factor and the FIT for each relative period. As seen in **Figure 3**, The forecast for 2024 suggests that the trend in sales will continue to increase. The highlighted section is the new forecasted demand for 2024 and every month's sales are higher than those in 2023. One noticeable trend in 2024 is that September has a higher forecasted trend than any of the other years, indicating positive growth for that month. In **Figure 5**, through analyzing several key factors, we see that the chart shows an upward trend in sales over a period of time that includes 2024. To find the underlying trend, data was deseasonalized which removes the effect of seasonality. This showed in **Figure 5** that there is still a steady increase in sales without the effect of future seasonality.

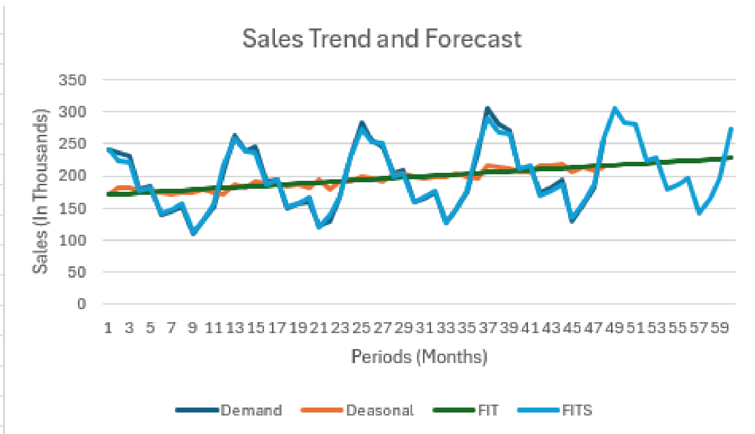

Figure 5

Time Series Adjusting Forecasting

When forecasting data for sales in 2024, it is important to be very accurate. Using time series, MSE, MAD, and MAPE are integrated to measure differences in between forecasted values and actual data. MAD measures the average absolute differences between actual sales and forecasted sales. The MAD for Joe's Stone Crab was 4.34, as seen in **Figure 4**. This MAD suggests that the model is quite accurate with only minor deviations from the actual sales. MSE measures the average of the squared differences between the actual and forecasted values. With an MSE of 32.15,

MAD	4.34
MSE	32.15
MAPE	2.24%

Figure 4

the squared errors are relatively low. The MAPE measures the accuracy of the forecast as a percentage of the actual sales. In Figure 4, we have a MAPE percentage of 2.24 which suggests an accurate forecast, as the percentage error is very low. The way we look at it is that the lower the error is with the forecasted and actual data, the more accurate the model is.

Business Recommendations

While Joe’s Stone Crab is a successful business, there are things that they can still improve on. Based on **Figure 1**, sales seem to consistently be higher in the winter months like December, January, February, and even up to March. To ensure that Joe’s Stone Crab can keep up with the increasing demand, they should increase the stock of menu items and the amount of staff working. This would hopefully help in maximizing revenue during their peak periods. Another recommendation would be to have promotions during periods that are known to be slower than others. This would most notably include September and October. Joe’s Stone Crab should introduce discount meal packages, special Fall seafood offerings, and even a rewards system for customers. Another recommendation to add on is expanding its business. As previously seen in **Figure 5**, Joe's Stone Crab is increasing in sales each month of every year. This shows great potential growth for the business and how much they can benefit from their business. The company should increase its marketing sector to find new customer segments, like corporate clients. Also, the company should consider opening their restaurant in several locations to get their services noticed by more people throughout the country. With their business already blowing up in one location, there is a good chance that opening in different locations will do the same. Along with different locations, the owners of the restaurant can start partnerships to fully capitalize on increasing customer demand. The final recommendation I have for Joe's Stone Crab is to use forecast data for any decision making. The errors that were provided in **Figure 5** shows that the forecasts are reliable and accurate in finding future sales and trends. This data can be used in any significant way like financial planning or inventory management.

Conclusion

The analysis of Joe’s Stone Crab has proved that the forecast model is reliable and that Joe’s Stone Crab growth trends are increasing each year. The forecast for 2024 suggests that the company will continue to grow, especially during peak months, which will definitely increase profitability for the business. The forecast model’s low error values from the MAD, MSE, and MAPE support that the company is expected to grow in the future considering the historical data. A few things that the Joe’s Stone Crab can do to maximize profits is to possibly expand the business into several locations to be more well known. Another task they could focus on doing is attracting customers to increase sales in less busy months. By creating an accurate forecast model, Joe’s Stone Crab can make smarter decisions about future inventory and marketing to increase sales at an even faster rate and increase profitability.